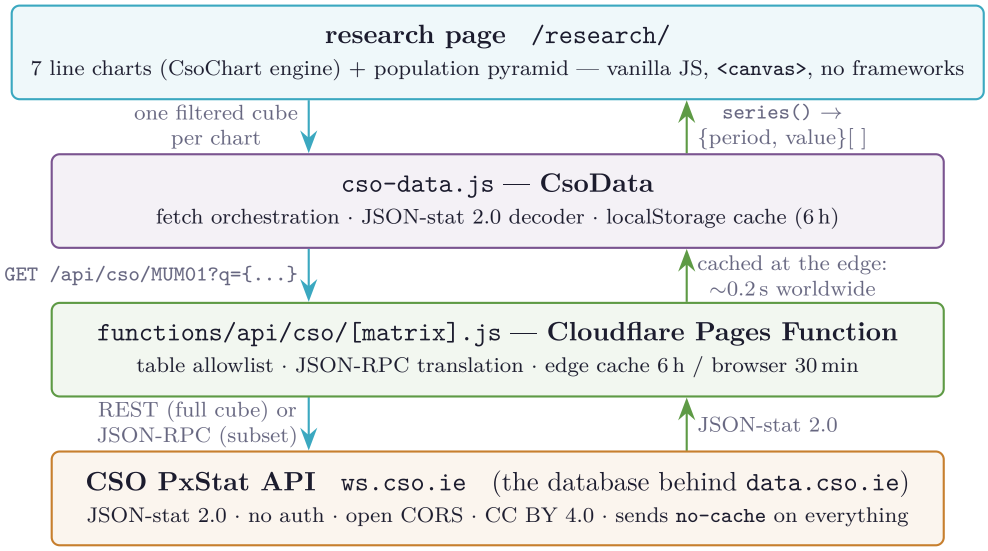
Icecon.ie × CSO PxStat

The live data pipeline, at a glance

July 2026 — every fact verified against the live API

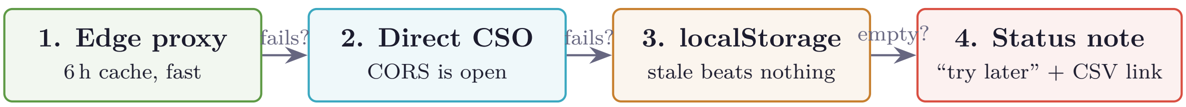
1 The whole system



Deploy is *push to main*: Cloudflare Pages rebuilds the static site and the proxy function together, ∼1 minute, zero configuration.

2 Why a proxy? The four-level fallback

The CSO sends `cache-control: no-cache` on every response, so without our own caching every student pays a cold round trip to CSO origin. The page also never breaks: each level below catches the failure of the one above.



Level 2 also covers local development (`python3 -m http.server` runs no Pages Functions), so the same code works everywhere unchanged.

3 Decoding JSON-stat: the one piece of real arithmetic

A cube arrives as one flat `value[]` array plus its dimensions. The flat index is row-major over the `id` order — exactly like nested loops:

```
id   = [STATISTIC, TLIST(M1), Age, Sex]
size = [    2    ,    342   ,  3 ,  3 ]
```

↓

strides (right to left): `[3078, 9, 3, 1]`
`value[stat·3078 + month·9 + age·3 + sex]`

↓

example — rate (%), June 2026, 15–74, both sexes:
`value[1·3078 + 341·9 + 1·3 + 0] = value[6150] = 4.1`

`CsoData.series(ds, fixed)` does this for a whole time axis: pin every non-time dimension, walk the time categories, return `[{code, label, value}]`. One PxStat quirk: `category.index` is an *array* of codes (Eurostat uses an object) — the decoder handles both.

4 Anatomy of a chart block

• All items • Housing & energy ○ Transport ○ Food & drink ...	**series pills**: multi-select overlay; colours = legend
[Annual inflation (%)] [Index (Dec 2023=100)]	**unit toggle**: same cube, no refetch
[View data table] [Dark ∨] [Download PNG] [Download CSV]	**actions**: 1600×900 themed PNG; CSO's own CSV
canvas: multi-series lines, hover crosshair + tooltip, zero line, year ticks	**canvas**: manim palette, glow on dark
Source: CSO PxStat **CPM20** · Updated 9 Jul 2026 · CC BY 4.0 · Cite as: *CSO, ...*	**citation**: auto-dated from the API

The engine builds all of this from a ∼20-line config; the pill list comes from live metadata, so new CSO categories appear automatically.

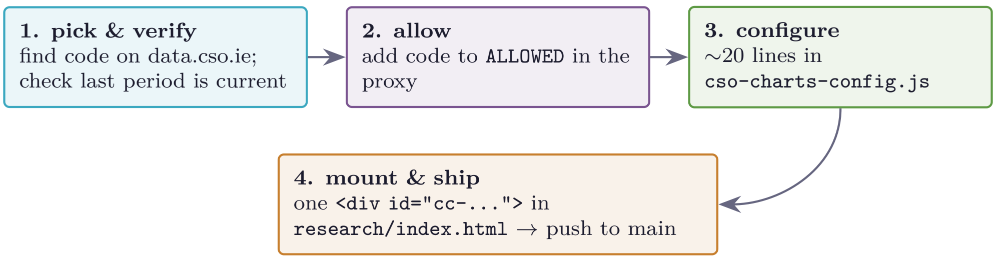
5 The visualisation suite and its tables

Three renderers share one data layer: **lines** (CsoChart), **donuts** with a year slider (CsoPie), and the **regional map** (CsoMap). A third build round added twenty more visuals — BoP in depth (BPQ15), reserve assets (BPQ18), UK trade and FDI after Brexit (BPQ35/38), energy dependence and demand (SEI01/06, TSM10, MEC02), aircraft leasing (ALIA01/03/09) and the international investment position and external debt (BPQ26/24) — with written analysis blocks on the page. The donut engine gained `annualize` (quarterly/monthly → annual sums) and `signed` (magnitude slices, sign in legend) for balance-of-payments data. Full inventory: `cso-data-pipeline.md`. The original charts:

Chart	Table	Span	Pills / toggle / slider
Unemployment	MUM01	1998– (monthly)	3 ages / % vs ’000s
Inflation (CPI)	CPM20	1996– (monthly)	14 commodity groups / % vs index
GDP–GNP–GNI–GNI*	NA001	1995– (annual)	4 aggregates, €bn
House prices	HPM09	2005– (monthly)	6 regions/types / index vs %
Dwelling completions	NDQ01	2011– (quarterly)	4 house types / raw vs SA
Government debt	GFQ12	2000– (quarterly)	single series, % of GDP
Merchandise trade	TSM01	1970– (monthly)	exports, imports, surplus (±SA)
Population pyramid	PEA11	1926– (year slider)	single year of age × sex
Tax revenue (donut)	ITXS01	1995– (year slider)	top-9 tax heads + Other
Gov. rev/exp/deficit	GFA01	1995– (annual)	revenue, expenditure, B9
Social protection (donut)	SPEA02	2000– (year slider)	9 benefit functions
Balance of payments	BPQ15	2002– (quarterly)	current acct, goods, services, income
Trade in services	BPA04	2003– (annual)	8 components / exports vs imports
Industrial production	MIM05	2015– (monthly)	modern vs traditional sector
Retail sales	RSM08	2021– (monthly)	6 business types / volume vs value
Services index	MSI03	2021– (monthly)	6 sectors / volume vs value
Housing map	NDQ08	2011– (map + slider)	8 NUTS3 regions, GISCO geometry

Trap to remember: the PxStat catalogue keeps discontinued tables with no flag (CPM13 ends 2016, HPM01 ends 2019, . . .). Before adopting any table, check the last period in its time dimension via `ReadMetadata`.

6 Adding chart #9



No drawing code, no fetch code, no DOM code — the engine owns all of it. Full prose reference: `docs/cso-data-pipeline.md`.